AEQUITAS

Decentralized Human Monetary System

"Money exists because people exist. Nothing more, nothing less."

Whitepaper v0.9 | June 2026

✓ Smart Contract deployed (Sepolia)	✓ Groth16 ZKP on-chain
✓ Layer 1 Blockchain (Go)	✓ BlockDAG Consensus
✓ P2P Network (2 Nodes)	✓ Android App (Fingerprint)
✓ Proof Server live	✓ Block Explorer live

1. The Problem

The global financial system is structurally unjust. Central banks control the money supply without democratic mandate. The US dollar as world reserve currency means the Federal Reserve conducts monetary policy for the entire world — without legitimacy. When the Fed raises rates, currencies collapse in Pakistan, Egypt and Nigeria. The person in Zimbabwe or Venezuela made no choice. They were born into a system that fails — and cannot even buy dollars because their passport is the wrong one.

The digital euro exacerbates this problem: programmable money with expiration dates, full transaction visibility for authorities, spending that can be tied to conditions. Technologically, this is the maximum opposite of financial freedom.

2. The Solution: Aequitas

Aequitas is a decentralized monetary system that ties money supply directly to verified human existence. No state, no bank, no committee can arbitrarily expand the money supply. The code decides — automatically, transparently, immutably.

Core Principle:

```
Total Supply = Verified Humans x 1,000 AEQ
```

3. Technical Architecture

3.1 Layer 1 Blockchain (BlockDAG)

Aequitas operates its own Layer 1 blockchain in Go with BlockDAG consensus. Unlike linear blockchains (Bitcoin, Ethereum), BlockDAG allows multiple blocks to be produced in parallel and merged into a single block with multiple parents. This enables high throughput without sacrificing security — similar to Kaspa.

BlockDAG Structure:

```
Block #154 (MERGE) Parent 1: a3f8c2... (Node 1 - Railway) Parent 2: 9d2e1f...
(Node 2 - Render)
```

Live Infrastructure:

Component	URL / Address
Block Explorer	aequitas-production-9fba.up.railway.app
Bootstrap Node	thomas.proxy.rlwy.net:47298
Node 2	aequitas-node-2.onrender.com
Proof Server	aequitas-proof-server-production.up.railway.app
Smart Contract (V5)	0x4f147d5B3388AF07993CC4fC548502A78Af0B8b5

3.2 Proof of Humanity (Biometric ZKP)

Registration requires a Groth16 Zero-Knowledge Proof of biometric uniqueness. Biometric data never leaves the device. Only the mathematical proof is submitted on-chain.

ZKP Commitment Formula:

```
commitment = ZKP( biometricHash x walletAddress + deviceSalt )
```

Phase 1 (upcoming): PPG cardiac biometrics via MAX30102 sensor — believed to be the world's first application of PPG-based biometrics for decentralized identity. Finger on camera + flash: stable PPG signal without external device, device-independent, free.

3.3 Aequitas Index — Algorithmic Monetary Policy

The Aequitas Index automatically measures economic health and adjusts monetary policy without human intervention. No committee, no political decisions.

Index Formula:

```
Index = Velocity x 40% + Growth x 35% + (100 - Gini) x 25% Index < 40 ->
Inflation (0-1.5% annual, equal distribution) Index 40-60 -> Neutral Index >
60 -> Wealth Cap active, overflow redistributed equally
```

4. Economic Model

Parameter	Value
Initial Grant	1,000 AEQ per verified human
Supply Cap	Humans x 1,000 AEQ (dynamic)
Transaction Fee	0.1%
Fee Distribution	40% Validators / 30% LPs / 20% UBI / 10% Treasury
Max Inflation	1.5% p.a. (algorithmic, from 10,000 humans)
Wealth Cap (Phase 3)	5x fairShare
Block Time	6 seconds
Consensus	Proof of Humanity + BlockDAG

Wealth Cap Phases:

Phase	Registrations	Max Balance
0	0-100	No cap
1	100-1,000	20x fairShare
2	1,000-10,000	10x fairShare
3	10,000-100,000	5x fairShare
4	100,000+	Gini-dynamic 3-5x

5. Current Status (June 2026)

✓ Smart Contract AequitasV5 deployed on Sepolia

✓ Groth16 ZKP biometrics verified on-chain

✓ ERC-20 Token AEQ with full economic model

✓ 0.1% transaction fee with automatic pool distribution

✓ Layer 1 Blockchain in Go (BlockDAG)

✓ P2P Network with 2 active nodes (Railway + Render)

✓ Real BlockDAG merge proven (multiple parent blocks confirmed)

✓ Block Explorer live with DAG visualization

✓ Android App with Hardware Secure Element (fingerprint)

✓ Proof Server with Sybil protection

✓ Keeper Bot (hourly economic cycles)

✓ 4 verified humans in the system

6. Roadmap

Phase	Status	Milestones
Phase 0	Complete	Smart Contracts, ZKP, Android App, Proof Server, Keeper Bot
Phase 0+	Complete	Layer 1 Chain, BlockDAG, P2P Network, Block Explorer
Phase 1	Ongoing	MAX30102 PPG Sensor, iOS App, Grant Application
Phase 2	Planned	Public Testnet, Lending Protocol, DEX
Phase 3	Planned	Mainnet Launch, Cross-Chain Bridges

7. Open Research Questions

Aequitas is honest about its open questions. These are not weaknesses — they are the research agenda for the next phase:

• Device-independent biometric verification: PPG via camera vs. dedicated sensor

• Optimal calibration of the Fuzzy Commitment Circuit for different cameras

• Long-term stability of the Aequitas Index with growing user base

• Governance mechanism for code updates without central authority

• Scaling BlockDAG with thousands of parallel nodes

"Money exists because people exist. Nothing more, nothing less."

github.com/hanoi96international-gif/Aequitas
aequitas-production-9fba.up.railway.app